

Iris Sun

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Full Stack Java Developer

Professional Synopsis

- **6+ years of professional experience** in software engineering with a strong background in full-stack development, data engineering, and cloud-native applications across diverse technology stacks.
- Strong foundation in object-oriented programming with hands-on experience across **Java, Python, Kotlin, JavaScript, and SQL**, developing scalable backend services and interactive front-end applications.
- Proficient in modern web development using **React, AngularJS**, and **Node.js**, coupled with API development and integration using **JSON, HTML, CSS**, and **Postman**.
- Experienced with **Spring Boot** and **JPA** for building robust, secure Java-based microservices, including RESTful APIs and database transaction management.
- Applied **data science and machine learning techniques** using **Python**, leveraging libraries such as **Scikit-Learn, NumPy, Pandas, and XG-Boost** for data preprocessing, model training, and evaluation.
- Skilled in **SQL, Hive**, and **MongoDB** for data querying and transformation across relational and NoSQL environments; familiar with large-scale data processing using **Apache Spark**.
- Maintained CI/CD pipelines with **Jenkins** and Docker, automating build, test, and deployment cycles across multiple environments.
- Implemented real-time data pipelines with **Apache Kafka**, building robust consumers/producers and managing partitioning, offsets, and topic retention policies.
- Built automated testing pipelines using **JUnit, Mockito**, and **Selenium**, achieving >90% unit test coverage and maintaining long-term code quality.
- Familiar with cloud platforms including **AWS** (EC2, Lambda, S3, RDS, IAM, CloudWatch) and **GCP** (Compute Engine, Cloud Storage, Pub/Sub), with experience in deploying and monitoring applications.
- Capable of building interactive dashboards and reports using **Tableau** and **PowerBI** to drive data-informed decision-making across business units.
- Built secure authentication and authorization workflows using **Spring Security** and JWT-based access control for sensitive endpoints.
- Familiar with building CI/CD pipelines and deployment scripts for WAR/JAR packaging using **Ant, Maven**, and **Gradle** in production environments.
- Maintained version control and collaborative development using **Git, Bitbucket**, and feature branching strategies across large codebases.
- Demonstrated ability to independently deliver features end-to-end, collaborate in agile teams, mentor junior developers, and communicate effectively with cross-functional stakeholders.

- Documented systems and processes using **LaTeX** and Markdown, delivering high-quality reports, design specs, and technical documentation for internal use and hand-offs.

Technical Skills

- **Programming Languages:** Java, Python, Kotlin, JavaScript, SQL, Bash, R, MATLAB, HTML, CSS, LaTeX, JSON
- **Frameworks & Libraries:** Spring Boot, Spring Data JPA, Spring Security, JPA, React, AngularJS, Node.js, Scikit-Learn, XGBoost, Pandas, NumPy
- **Databases:** MySQL, PostgreSQL, MongoDB, Hive, Redis
- **DevOps & Tools:** Docker, Kafka, Jenkins, Git, Gradle, Maven, Postman, IntelliJ, Eclipse, Mockito, PowerMock, JUnit
- **Cloud Platforms:** AWS (EC2, Lambda, S3, RDS, IAM, CloudWatch), GCP (Compute Engine, Cloud Storage, Pub/Sub)
- **Big Data & Streaming:** Apache Spark, Apache Hive, Apache Kafka
- **Data Visualization & BI:** Tableau, Power BI
- **Operating Systems:** Windows, macOS, Linux/Unix
- **Project & Workflow Management:** JIRA, Agile (Scrum/Kanban)

Professional Experience

PetSmart

Sep 2024 – present

Software Engineer

Seattle, WA

Project Description:

PetSmart is the largest specialty pet retailer in North America, and I worked on the **E-commerce Logistics Platform team**, leading the modernization of backend systems that manage real-time inventory, checkout stability, and multi-channel order orchestration. The project involved decomposing a monolithic architecture into Spring Boot microservices, implementing event-driven pipelines with Kafka, and enhancing system resilience through circuit-breaking, observability, and disaster recovery strategies. Our work enabled scalable, fault-tolerant order processing and inventory synchronization across digital and physical retail channels.

Roles & Responsibilities:

- **Led the architecture redesign** of the inventory tracking module, migrating a monolithic Java system into distributed **Spring Boot microservices**, reducing deployment time by **60%** and significantly improving code modularity.
- **Designed and implemented a GraphQL gateway layer** to unify inventory access across web, mobile, and in-store platforms, decreasing redundant data fetching by **45%**.
- Refactored legacy modules into **asynchronous workflows** using **Kafka** and **Redis queues**, improving order processing throughput by **25%** under peak load.
- **Engineered graceful degradation strategies** with **Resilience4j**, allowing critical services like checkout to remain operational even during partial backend outages.
- Implemented **domain-driven design (DDD)** to define clean service boundaries between inventory, order management, and logistics, improving system clarity and testability.
- Conducted **failover simulation drills** and co-authored disaster recovery runbooks, helping ensure **99.99% checkout availability** during regional outages.
- Integrated centralized exception tracing with **Spring AOP**, **Prometheus**, and **Grafana**, cutting incident resolution time from 40 to 15 minutes on average.
- **Refactored backend concurrency logic using Kotlin's coroutines**, reducing null-safety exceptions and simplifying reactive data flows.
- **Deployed containerized services to AWS EKS**, configured durable storage with **S3**, and monitored health metrics and logs with **CloudWatch**.
- Designed Redis-backed **token bucket rate limiting** for inventory lookups, reducing service stress without degrading user experience.
- Tuned complex **SQL and JPQL queries** using execution plans and caching hints, improving inventory lookup performance by **40%**.
- Created and consumed resilient service clients using **Feign** and **RestTemplate**, applying timeout and fallback logic to mitigate service dependency risks.
- Designed **denormalized MongoDB schemas** for high-volume event logs, boosting write throughput by **50%** during sync operations.
- Registered services using **Spring Cloud Netflix Eureka**, enabling dynamic scaling and service discovery across cloud environments.
- Worked closely with **mobile engineering** to deliver lightweight APIs for React Native clients, decreasing cart interaction latency by **35%**.
- Served as a **technical liaison** with the 3PL logistics vendor integration team, aligning contract schemas and SLA enforcement for next-day fulfillment services.
- Participated in **CI/CD automation** with Jenkins, implementing blue/green deployments and automatic rollback strategies for safer releases.
- Collaborated in Agile sprints, wrote detailed engineering documentation in **Confluence**, and actively contributed to retrospectives and design reviews.

Environment:

Java 11, Kotlin, Spring Boot, Spring Cloud, GraphQL, Kafka, Redis, MongoDB, MySQL, JPA, Feign, RestTemplate, Spring AOP, Resilience4j, Prometheus, Grafana, AWS EKS, AWS S3, AWS CloudWatch, Docker, Jenkins, JIRA, Confluence, Git

Chewy

Jul 2023 – May 2024

Software Engineer - Fulfillment & Inventory Platform

Seattle, WA

Project Description:

Chewy is a leading U.S. e-commerce company for pet supplies, and I worked on its Fulfillment & Inventory Platform team, which powers backend services for order processing, inventory tracking, and warehouse coordination. I developed microservices using Java, Spring Boot, and Kafka to synchronize real-time inventory and order state across systems, implemented Redis for low-latency caching, and deployed services via Docker and Kubernetes. Our work improved fulfillment accuracy, reduced inventory sync latency by 35%, and increased system resilience during peak seasonal traffic.

Roles & Responsibilities:

- Developed backend services in **Java** using **Spring Boot**, handling order orchestration, fulfillment tracking, and payment validation flows.
- Integrated **Kafka** to stream order and inventory events across microservices, enabling real-time synchronization between checkout, fulfillment, and customer notification systems.
- Designed RESTful APIs consumed by frontend and mobile teams, providing endpoints for cart operations, order status, and reordering history.
- Built and tuned Redis-backed caching layers to store active SKU inventory and order sessions, reducing database query volume by over 40%.
- Worked with **Kubernetes** and **Docker** to deploy and manage containerized microservices across staging and production environments.
- Configured Jenkins CI/CD pipelines to automate testing, packaging, and deployment across environments with rollback support.
- Applied **JUnit** and **Mockito** for unit and integration tests, increasing backend code coverage to 92% and reducing post-deployment incidents.
- Monitored system health with **Prometheus** and visualized application performance and traffic spikes with **Grafana** dashboards.
- Tuned SQL queries and added indexes to optimize warehouse lookup and product availability queries under peak load.
- Collaborated with product managers and UX engineers in Agile ceremonies to refine requirements, prioritize tech debt, and unblock delivery pipelines.

- Wrote batch jobs in **Kotlin** to process refund settlements and push nightly order analytics to the BI platform.
- Mentored two junior developers by reviewing code, pairing on service design, and guiding onboarding through architecture walkthroughs.

Environment:

Java, Spring Boot, Apache Kafka, Redis, MySQL, Docker, Kubernetes, Jenkins, JUnit, Mockito, Prometheus, Grafana, Kotlin, RESTful APIs, Agile

Ant Financial

Apr 2020 – Aug 2021

Software Engineer – Backend Systems

Hangzhou, China

Project Description:

At Ant Financial, I contributed to the backend development of the “My Assets” and transaction reconciliation modules for Alipay’s wealth management platform. This involved working with large-scale distributed systems and securing data transactions between internal services and third-party APIs. The role required building microservices for real-time fund availability, transaction summaries, and integrating analytics pipelines into Ant’s risk control platform.

Roles & Responsibilities:

- Developed and maintained Java-based microservices using **Spring Boot** and **Spring Cloud**, handling account summaries, fund availability, and audit trails.
- Designed and tuned **MySQL** and **Hive** queries to optimize batch reconciliation jobs that processed millions of transactions per day.
- Implemented **REST APIs** and **GraphQL** endpoints to serve aggregated transaction views across multiple asset types (funds, insurance, savings).
- Integrated **Kafka** for real-time stream processing of investment activity, enabling fraud detection and transaction classification.
- Used **Spring AOP** to introduce cross-cutting logic for transaction logging, data masking, and error tracing.
- Wrote unit and component tests using **JUnit** and **Mockito**, achieving >90% test coverage for critical services.
- Created Swagger-enabled documentation for all internal APIs and tested endpoints using **Postman**.

- Containerized services using **Docker**, and deployed them via **Aliyun ECS** using company-wide DevOps CI/CD platform.
- Worked closely with data science team to expose data aggregation APIs used in modeling risk and customer segmentation.
- Participated in weekly Agile sprints, reviewed code via Bitbucket, and documented tech designs using Confluence and Markdown.
- Refactored legacy Java services to adopt **Java 11** and functional paradigms such as lambdas and streams, reducing LOC and improving readability.
- Actively contributed to database migration planning and rollback logic during system switchover phases.

Environment:

Java, Spring Boot, Spring Cloud, MySQL, Hive, Kafka, Docker, Aliyun ECS, Redis, Gitee, Swagger (local), JUnit, Mockito, REST, AOP, Postman, Agile

Costco Wholesale

Nov 2018 –Feb 2020

Software Engineer – Inventory & Pricing Systems

Seattle, WA

Project Description:

At Costco, I was part of the backend engineering team responsible for modernizing the inventory and pricing system used across the company's warehouse and e-commerce platforms. The initiative involved migrating legacy services to microservices, integrating real-time pricing logic, and enhancing data consistency between supply chain, ERP, and POS systems.

Roles & Responsibilities:

- Designed and developed Java-based microservices using **Spring Boot** to manage product catalog operations including SKU creation, lifecycle state, and promotional rules.
- **Led Redis-based pricing engine** handling dynamic regional pricing for over **15K SKUs/min** with **99.95% accuracy**, reducing pricing discrepancies across e-commerce and POS systems by 80%.
- Integrated **Kafka** to distribute product and pricing updates across downstream systems (checkout, analytics, and warehouse sync), achieving real-time propagation under 1.5s end-to-end latency.
- Spearheaded the **refactor of legacy Oracle-based pricing logic** into modular PostgreSQL services with versioned schema migrations, enabling faster iterations and rollback-safe deployments.

- Acted as **technical lead** during the integration of ERP inventory APIs, coordinating efforts across warehouse, finance, and online retail teams to ensure transactional consistency.
- Containerized core services using **Docker** and deployed via **Kubernetes (Rancher)** with blue-green strategy, enabling zero-downtime releases across three production environments.
- Implemented access control and audit traceability using **Spring Security** and JWT, satisfying internal audit and compliance mandates.
- Introduced **structured logging with correlation IDs** and centralized alerting through Grafana + Prometheus, reducing mean time to resolution (MTTR) by 60% during production incidents.
- Championed the adoption of **Resilience4j** for circuit-breaking and fallback handling, improving service uptime and isolating failures from downstream systems.
- Authored automated integration test suites with **JUnit**, **Mockito**, and **Postman**, increasing test coverage from 68% to 92% in critical modules.
- Partnered with BI and supply chain teams to push real-time inventory and price deltas into a **Kafka-based analytics pipeline** powering **Tableau** dashboards.
- Contributed to Costco's internal Java SDK for standardized logging, API response structures, and database transaction wrapping.
- Designed and executed a **load testing strategy using JMeter** and real production data patterns, identifying bottlenecks that led to a 35% improvement in service response times during seasonal traffic surges.
- Initiated and led a **cross-departmental initiative** to standardize product metadata APIs, aligning backend data contracts between pricing, inventory, and mobile checkout teams, reducing integration time for new features by 50%.

Environment:

Java, Spring Boot, Kafka, PostgreSQL, Redis, Docker, Kubernetes, Rancher, JUnit, Mockito, Grafana, Tableau, REST APIs, Agile, Spring Security

Education

University of Washington

Seattle, WA

Master of Science in Information Systems

University of Washington

Seattle, WA

Bachelor of Science